MUHAMMAD ABUBAKAR

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ABOUT ME

AI Engineer specializing in agentic workflows, LLM systems, and end-to-end AI deployment. Experienced in building production-ready RAG pipelines, multi-agent systems, and scalable backend architectures. Strong background in machine learning, deep learning, and AI system design, with hands-on expertise in LangChain, vector databases, and LLM fine-tuning.

SKILLS

Languages: Python
AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face
LLM & GenAI: RAG, OpenAI API, Gemini API, Fine-Tuning (SFT, DPO, LoRA, QLoRA)
Agentic Frameworks: LangChain, CrewAI
Vector DBs: Pinecone
Backend: FastAPI, REST APIs

PROJECTS

End-to-End RAG Pipeline (n8n + Pinecone + LLM)

- Developed a production-ready RAG system with document ingestion, embedding, and retrieval
- Used Pinecone vector database for semantic search and context retrieval
- Implemented context aware response generation, improving answer accuracy significantly

LLM Fine-Tuning (LLaMA – Marketing Domain)

- Performed supervised fine-tuning + DPO (Direct Preference Optimization)
- Built instructional datasets with accepted/rejected responses
- Enhanced model performance via hyperparameter tuning and dataset curation

AI-Powered Customer Support System (FastAPI + LLMs)

- Built a chat-based AI system handling customer queries and order management
- Integrated LLM APIs with structured backend logic
- Designed dynamic retrieval system using PostgreSQL DataBase
- Developed scalable backend using FastAPI architecture

EDUCATION

2021 - 2025
B.Sc. Mechanical Engineering, University of Engineering and Technology, Lahore

2019 - 2021
F.Sc. Pre-Engineering, Govt. Islamia College Civil Lines, Lahore

CERTIFICATIONS

Machine Learning Specialization (Stanford University & DeepLearning.AI)
https://coursera.org/share/8478f2029ea1d4066f9a140cbd7c9e04